# hoopR :: CHEAT SHEET

Men's basketball data in R: NBA, NBA G-League, men's college basketball and 23 international league families plus EuroLeague / EuroCup. Tidy play-by-play, box scores, schedules and rosters from ESPN, a full `stats.nba.com` wrapper, KenPom / Torvik / CollegeBasketballData scrapers, Basketball-Reference, Fox and RealGM, plus season-level loaders from the SportsDataverse data releases. Women's sibling: **wehoop**; Python mirror: `sportsdataverse.nba` / `.mbb` in **sportsdataverse-py**. Page 1 of 4: install, surfaces, loaders, credentials, errors, helpers.

v3.2.0

R >= 4.1.0 · httr2 · data.table
hoopR.sportsdataverse.org

## GET STARTED

```
install.packages("hoopR")              # CRAN
pak::pak("sportsdataverse/hoopR")      # main
remotes::install_github(
  "sportsdataverse/hoopR")
library(hoopR)
```

1,316 exports (667 in the international families). Every function returns a tibble / data.frame (or a named list of tibbles for multi-table endpoints). Prefix tells you the source; page tells you where to look.

## SURFACES BY PREFIX

| Prefix | Source | Page |
|---|---|---|
| `load_nba_* load_mbb_* load_nba_stats_* load_ncaa_mbb_*` | season parquet/rds releases (zero scraping) | 2 · 3 |
| `espn_nba_* espn_mbb_*` | ESPN Site / Core APIs, live | 2 · 3 |
| `nba_*` | stats.nba.com Stats API (~160 endpoints) | 4 |
| `nbagl_*` | G-League (Stats API, league 20) | 3 |
| `nba_l2m* nba_referee_assignments` | official.nba.com L2M + referee crews | 3 |
| `euroleague_*` | EuroLeague / EuroCup APIs (keyless) | 3 |
| `fiba_* genius_* atrium_* acb_* kbl_* ...` | 23 international families + 148 competition aliases | 3 |
| `kp_*` | KenPom (subscription login) | 2 |
| `cbbd_*` | CollegeBasketballData API (key) | 2 |
| `torvik_*` | Bart Torvik T-Rank | 2 |
| `ncaa_mbb_*` | NCAA.com NET + teams | 2 |
| `fox_nba_* fox_mbb_*` | Fox Sports (Bifrost) | 2 · 3 |
| `bref_*` | Basketball-Reference | 3 |
| `realgm_*` | RealGM (needs chromote + Chrome) | 3 |
| `spotrac_ hoopshype_ nbadraft_ rotowire_` | salaries · mock draft · injuries | 3 |
| `*_crosswalk` | ESPN ⇄ NBA Stats / KenPom / Torvik / Fox ids · MBB team names | 2 · 3 |

## DATA LOADERS (FAST PATH)

Full seasons from the `sportsdataverse-data` GitHub releases; each loader takes exactly the seasons its release publishes. Seasons are the **end year** (2024 = 2023-24). ESPN join ids load as integer.

```
pbp  <- load_nba_pbp(seasons = 2022:2024)
tb   <- load_nba_team_box(
          seasons = most_recent_nba_season())
pb   <- load_mbb_player_box(seasons = 2024)
sch  <- load_mbb_schedule(seasons = 2020:2024)
# stream straight into a DB table
load_nba_pbp(2024, dbConnection = con,
             tablename = "nba_pbp")
```

| Dataset | `load_nba_*` | `load_mbb_*` |
|---|---|---|
| play-by-play | `pbp` | `pbp` |
| box scores | `team_box player_box` | `team_box player_box` |
| schedules | `schedule` | `schedule` |
| standings | `standings` | `standings` |
| rosters | `rosters game_rosters` | `rosters game_rosters` |
| season stats | `player_stats team_stats` | `player_stats team_stats` |
| people | `player_core officials draft` | `player_core officials` |
| crosswalks | `team_ schedule_ player_crosswalk` | same three |
| shots / value | `shots player_impact` | `shots ratings player_value` |
| conferences | `groups group_seasons group_aliases team_group_seasons` | same four |

Also `load_nba_stats_*` — 18 stats.nba.com-derived release loaders (schedule pbp possessions lineups shots rosters boxscores game_logs season_stats leaguedash standings officials coaches draft) + `update_nba_stats_db()` — and `load_ncaa_mbb_*` — 13 stats.ncaa.org loaders incl. `rapm` / `rapm_within_team`. Group loaders keep each season's own conference alignment.

**Local play-by-play database**

```
update_nba_db(dbdir = ".",
              dbname = "hoopR_db",
              tblname = "hoopR_nba_pbp")
update_mbb_db()    # RSQLite / DBI (Suggests)
con <- DBI::dbConnect(RSQLite::SQLite(),
                      "hoopR_db")
```

## LIVE ESPN + STATS API

```
# ESPN: game_id from espn_*_scoreboard
#       or from load_*_schedule
espn_nba_scoreboard(20230423)   # YYYYMMDD
espn_nba_pbp(game_id = 401071880)
espn_mbb_game_all(401479672)    # pbp + boxes
# stats.nba.com: 10-char game ids, "2023-24"
nba_pbp("0022200021", on_court = TRUE)
nba_leaguedashplayerstats(
  season = year_to_season(2023))
```

ESPN wrappers take `game_id`, `team_id`, `athlete_id`, `season` / `year`. Stats API wrappers mirror the endpoint's query params (snake_case) and return a **named list** of result sets. Live `espn_*_pbp` plays carry the releases' clock, period and timeout columns.

## CREDENTIALS & OPTIONS

```
# KenPom (active subscription)  ?kp_user_pw
Sys.setenv(KP_USER = "you@example.com",
           KP_PW = "...")
has_kp_user_and_pw()   # TRUE before kp_*()
# CollegeBasketballData  ?register_cbbd
Sys.setenv(CBBD_API_KEY = "XXXX-YOUR-KEY")
has_cbbd_key(); cbbd_key()
# or persist both in ~/.Renviron
usethis::edit_r_environ()
# session-wide proxy (stats.nba.com / ESPN)
options(hoopR.proxy = "http://host:port")
options(hoopR.fox_data_key = "...")  # Fox
```

Stats API requests go through `request_with_proxy()`, which sets the required `x-nba-stats-*` / Referer headers; `hoopR.proxy`, `http_proxy` env or an explicit `proxy=` arg are honoured. `progressively()` + `progressr` give progress bars on multi-season loads.

## ERRORS & DEFAULTS

```
r <- tryCatch(acb_pbp(match_id = 105401),
  hoopR_no_data = function(e) NULL)
# 404, nothing there  -> hoopR_no_data
# request failed      -> hoopR_asset_fetch
# 400 / 422           -> hoopR_bad_request
```

International, EuroLeague and officiating wrappers raise classed conditions (all inherit `hoopR_error`; EuroLeague and officiating use `hoopR_fetch_error`, EuroLeague `hoopR_invalid_request`) instead of returning an empty frame. Stats API `season` now defaults to the previous season, `year_to_season(most_recent_nba_season() - 2)`: pass it for the current one.

## HELPERS

| | |
|---|---|
| `most_recent_nba_season()` | current NBA season end year |
| `most_recent_mbb_season()` | current college season end year |
| `year_to_season(2023)` | `"2023-24"` for Stats API |
| `csv_from_url(...)` | fast csv reader behind loaders |
| `progressively(f, p)` | wrap `f` with a progressor |
| `login()` | KenPom session (reads `KP_USER` / `KP_PW`) |
| `hoopR::: request_with_proxy()` | httr2 GET w/ NBA headers + proxy (internal) |
| `hoopR:::rds_from_url()` `:::check_status()` | internal readers / status guard |
| `teams_links` | bundled KenPom team + conference names by year (2002-27, 9,044 rows) |
| `mbb_team_names` | NCAA / 247 / Sports-Reference / Warren Nolan names, 368 MBB teams |
| `parameter_descriptions` | all 300 Stats API params, values + defaults |
| `nba_player_dict()` `nba_team_dict()` | Stats-API id dictionaries |
| `nba_player_headshot_url()` `nba_team_logo_url()` | CDN media URLs |

## OVERVIEW TOPICS (?NAME)

`?hoopR_loaders` · `?espn_basketball` · `?nba_stats` · `?kp` · `?register_cbbd` · `?espn_basketball_player_core`. pkgdown reference groups follow the same split.

## WIDER SPORTSDATAVERSE

| | |
|---|---|
| **wehoop** | WNBA + women's college, same API shape |
| **bigballR** | stats.ncaa.org MBB lineups / on-off |
| **cfbfastR · baseballr · fastRhockey** | sister sport packages |
| **sportsdataverse-py** | `sportsdataverse.nba` (`nba_stats_*`, `espn_nba_*`, `load_nba_*`) · `sportsdataverse.mbb` (`espn_mbb_*`, `ncaa_mbb_*`) |
| **nba_api** | Python stats.nba.com reference |
| data repos | `sportsdataverse-data` (loaders) · `hoopR-nba-data` · `hoopR-mbb-data` · `hoopR-nba-stats-data` |

# Men's college basketball :: loaders · ESPN · KenPom · CBD · Torvik

Women's mirror: **wehoop** (`espn_wbb_*`, `load_wbb_*`) · stats.ncaa.org lineups: **bigballR** · Python: `sportsdataverse.mbb`

## LOADERS + ESPN · ESPN_MBB_*

### SEASON LOADERS

```
library(hoopR)
pbp <- load_mbb_pbp(seasons = 2022:2024)
tb  <- load_mbb_team_box(seasons = 2024)
pb  <- load_mbb_player_box(seasons = 2024)
sch <- load_mbb_schedule(seasons = 2006:2024)
load_mbb_standings(); load_mbb_rosters()
load_mbb_player_stats(); load_mbb_team_stats()
load_mbb_ratings(); load_mbb_shots()
load_mbb_player_value()
load_ncaa_mbb_{pbp,possessions,lineups,shots,
  rapm,team_box,player_box,rosters,...}()
# ^ stats.ncaa.org releases
update_mbb_db(dbdir = ".",
              tblname = "hoopR_mbb_pbp")
```

Releases: `espn_mens_college_basketball_*` tags on `sportsdataverse-data`; pbp / shots 2003 and 2006+, rosters 2025+.

### GAMES (LIVE ESPN)

```
espn_mbb_scoreboard("20221117")   # day slate
espn_mbb_pbp(game_id = 401479672)
espn_mbb_team_box(401479672)
espn_mbb_player_box(401479672)
espn_mbb_game_all(401479672)   # pbp + boxes
espn_mbb_wp(401256760)   # win prob per play
espn_mbb_betting(401256760)
espn_mbb_game_rosters(401256760)
```

Core-API per-game extras (event id): `espn_mbb_game_odds` `_game_predictor` `_game_probabilities` `_game_propbets` `_game_officials` `_game_broadcasts` `_game_situation` `_game_play` `_game_play_personnel` `_game_team_statistics` `_game_team_leaders` `_game_team_linescores` `_game_team_score` `_game_team_roster` `_game_powerindex`.

### TEAMS, PLAYERS, LEAGUE

```
espn_mbb_teams(); espn_mbb_conferences()
espn_mbb_rankings(); espn_mbb_standings(2021)
espn_mbb_team_stats(150, year = 2020)
espn_mbb_player_stats(athlete_id = 4593919,
                      year = 2021)
espn_mbb_team_roster(150, season = 2024)
espn_mbb_team_schedule(150, season = 2024)
espn_mbb_player_gamelog(athlete_id, 2024)
```

Also: `espn_mbb_team_current_roster` `_team_record` `_team_injuries` `_team_leaders` `_team_news` `_team_season_statistics` `_player_overview` `_player_splits` `_player_career_stats` `_player_eventlog` `_player_info` `_athletes_index` `_leaders` `_season_leaders` `_injuries` `_news` `_futures` `_powerindex` (BPI) `_tournaments` `_week_rankings` `_season_groups` `_coaches` `_venues` `_calendar`.

## KENPOM · KP_* (subscription)

```
Sys.setenv(KP_USER = "you@x.com",
           KP_PW = "***")     # ?kp_user_pw
kp_pomeroy_ratings(2020, 2024)   # years
kp_efficiency(2020, 2024)
kp_fourfactors(2020, 2024)
kp_team_schedule("Florida St.", year = 2022)
kp_team_players(team = "Duke", year = 2024)
kp_playerstats(metric = "eFG", year = 2024)
kp_box(game_id, year)
kp_winprob(game_id, year)
```

| Family | Functions |
|---|---|
| Ratings | `kp_pomeroy_ratings` `kp_pomeroy_archive_ratings` `kp_program_ratings` `kp_trends` `kp_hca` `kp_kpoy` |
| Team stats | `kp_efficiency` `kp_fourfactors` `kp_teamstats` `kp_pointdist` `kp_height` `kp_foul_trouble` `kp_playerstats` |
| Team pages | `kp_team_schedule` `kp_team_players` `kp_team_player_stats` `kp_team_history` `kp_team_lineups` `kp_team_depth_chart` `kp_gameplan` `kp_opptracker` `kp_minutes_matrix` `kp_coach_history` |
| Games | `kp_box` `kp_winprob` `kp_fanmatch(date)` `kp_game_attrs(year, attr)` |
| Conf / misc | `kp_conf(year, conf)` `kp_confhistory` `kp_confstats` `kp_arenas` `kp_officials` `kp_referee` `kp_player_career` |
| Auth | `login()` `has_kp_user_and_pw()` `kp_user_email()` `kp_password()` |

Team names must match KenPom spelling: `teams_links` lists them, with conferences, by year (2002-27). Team pages build their link from the name.

### BART TORVIK · TORVIK_*

```
torvik_ratings(year = 2024)   # T-Rank
torvik_team_factors(2024)
torvik_player_stats(2024)
torvik_game_schedule(2024)
torvik_game_stats(2024)
torvik_ncaa_results(2015, 2024, type = "team")
```

### CONFERENCE HISTORY (GROUPS)

```
load_mbb_team_group_seasons(2002:2025)
load_mbb_groups(); load_mbb_group_seasons()
load_mbb_group_aliases()  # ESPN/NCAA/KP
```

Membership as it was each season, not today's alignment; one `group_id` survives renames.

## COLLEGEBASKETBALLDATA · CBBD_*

```
# key: ?register_cbbd
Sys.setenv(CBBD_API_KEY = "XXXX")
cbbd_games(season = 2024, team = "Duke")
cbbd_plays_game(game_id,
  shooting_plays_only = FALSE)
cbbd_lines(season = 2024, conference = "ACC")
cbbd_ratings_adjusted(season = 2024)
cbbd_stats_player_season(2024, team = "Duke")
```

| Family | Functions |
|---|---|
| Games | `cbbd_games` `cbbd_games_teams` `cbbd_games_players` `cbbd_games_media` `cbbd_scoreboard` |
| Plays | `cbbd_plays_game` `cbbd_plays_date` `cbbd_plays_team` `cbbd_plays_player` `cbbd_plays_tournament` `cbbd_play_types` |
| Lineups / subs | `cbbd_lineups_game` `cbbd_lineups_team` `cbbd_substitutions_game` `cbbd_substitutions_team` `cbbd_substitutions_player` |
| Stats | `cbbd_stats_team_season` `cbbd_stats_team_shooting_season` `cbbd_stats_team_leaderboard` `cbbd_stats_player_season` `cbbd_stats_player_shooting_season` |
| Ratings / polls | `cbbd_ratings_adjusted` `cbbd_ratings_srs` `cbbd_ratings_elo` `cbbd_rankings` |
| Lines | `cbbd_lines` `cbbd_lines_providers` |
| Teams / venues | `cbbd_teams` `cbbd_teams_roster` `cbbd_teams_directory` `cbbd_teams_season_overview` `cbbd_conferences` `cbbd_conferences_history` `cbbd_venues` |
| Recruiting / draft | `cbbd_recruiting_players` `cbbd_recruiting_teams` `cbbd_recruiting_portal` `cbbd_draft_picks` `cbbd_draft_teams` `cbbd_draft_positions` |
| Key | `cbbd_key()` `has_cbbd_key()` `check_cbbd_key()` |

Most take `season`, `team`, `conference`, `start_date_range` / `end_date_range`, `season_type` filters.

## NCAA & FOX

```
ncaa_mbb_NET_rankings()   # current NET
ncaa_mbb_teams(year = 2024, division = 1)
fox_mbb_pbp(game_id)
fox_mbb_boxscore(game_id)
fox_mbb_odds(game_id); fox_mbb_teams()
fox_mbb_teams_all(); fox_mbb_league_leaders()
fox_mbb_team_roster(team_id)
fox_mbb_team_stats(team_id)
fox_mbb_team_gamelog(team_id)
fox_mbb_standings(team_id)
```

Fox ids differ from ESPN ids — bridge with the crosswalks below.

### CROSSWALKS (KEYED ON ESPN_TEAM_ID)

```
mbb_team_crosswalk(season = 2024)
# ESPN · KenPom · Torvik · Fox ids per team
mbb_schedule_crosswalk(season = 2024)
mbb_player_crosswalk(season = 2024)
load_mbb_team_crosswalk(2024)   # release
load_mbb_schedule_crosswalk()
load_mbb_player_crosswalk()
```

The team crosswalk adds six name columns (`ncaa_team` `espn_pbp_name` `warren_nolan_team` `team_247` `sref_slug` `sref_team`) from the bundled `mbb_team_names` (368 teams).

### TYPICAL PIPELINE

```
library(dplyr)
pbp <- load_mbb_pbp(2024)
shots <- pbp |>
  filter(shooting_play,
         !is.na(coordinate_x)) |>
  count(team_id, score_value)
kp <- kp_pomeroy_ratings(2024, 2024)
xw <- load_mbb_team_crosswalk(2024)
tb <- load_mbb_team_box(2024) |>
  left_join(xw,
    by = c("team_id" = "espn_team_id"))
```

Column names above (`shooting_play`, `coordinate_x`, `score_value`) follow the ESPN pbp schema documented in `?load_mbb_pbp`.

### ELSEWHERE

Per-game ESPN helpers used by the data repo builders: `helper_espn_mbb_pbp` `_team_box` `_player_box`. Python parity: `sportsdataverse.mbb.espn_mbb_pbp`, `load_mbb_pbp`, `ncaa_mbb_*` (bigballR port).

# NBA, G-League & international :: ESPN · EuroLeague · FIBA



## LOADERS + ESPN · ESPN_NBA_*

### SEASON LOADERS (1994+)

```
library(hoopR)
pbp <- load_nba_pbp(seasons = 2022:2024)
tb  <- load_nba_team_box(2024)
pb  <- load_nba_player_box(2024)
sch <- load_nba_schedule(2002:2024)
load_nba_standings(); load_nba_rosters()
load_nba_player_stats(); load_nba_team_stats()
load_nba_draft(); load_nba_player_core()
update_nba_db(dbdir = ".",
              tblname = "hoopR_nba_pbp")
```

Releases: `espn_nba_*` tags on `sportsdataverse-data`. Box scores, schedule, player stats / core 1994+; pbp, shots, standings 2002+; rosters 2025+.

### GAMES (LIVE ESPN)

```
espn_nba_scoreboard(20230423)   # YYYYMMDD
espn_nba_pbp(game_id = 401071880)
espn_nba_team_box(401071880)
espn_nba_player_box(401071880)
espn_nba_game_all(401283399)    # pbp + boxes
espn_nba_wp(401283399)
espn_nba_betting(401283399)
espn_nba_game_rosters(401283399)
```

Core-API per-event extras: `espn_nba_game_odds` `_game_predictor` `_game_probabilities` `_game_propbets` `_game_officials` `_game_broadcasts` `_game_situation` `_game_play` `_game_play_personnel` `_game_team_statistics` `_game_team_leaders` `_game_team_linescores` `_game_team_records` `_game_powerindex`. Live pbp carries the release's clock, period and timeout columns.

### TEAMS, PLAYERS, LEAGUE

```
espn_nba_teams(); espn_nba_standings(2021)
espn_nba_team_stats(team_id = 13, year = 2020)
espn_nba_player_stats(athlete_id = 1966,
                      year = 2022)
espn_nba_team_roster(13)
espn_nba_team_schedule(13, season = 2024)
espn_nba_team_depthchart(team_id = 13)
espn_nba_draft(season = 2024)
espn_nba_transactions(season = 2024)
espn_nba_freeagents(); espn_nba_injuries()
espn_nba_futures()
espn_nba_player_contracts(athlete_id = 1966)
espn_nba_powerindex(season = 2024)   # BPI
```

Also: `espn_nba_athletes_index` `_player_overview` `_player_gamelog` `_player_splits` `_player_career_stats` `_player_awards` `_player_contract` `_team_record` `_team_injuries` `_team_leaders` `_team_news` `_season_draft` `_draft_athletes` `_draft_rounds` `_coaches` `_franchises` `_venues` `_calendar` `_leaders` `_news`.

### G-LEAGUE · NBAGL_*

```
nbagl_schedule(season = year_to_season(2023))
nbagl_standings("2023-24")   # league 20
nbagl_players()
nbagl_pbp(game_id, on_court = TRUE)
nbagl_live_pbp(game_id)
nbagl_live_boxscore(game_id)
```

Same Stats-API backend as `nba_*` (page 4); most `nba_*` wrappers also accept `league_id = "20"`.

### FOX SPORTS · FOX_NBA_*

```
fox_nba_pbp(game_id)
fox_nba_boxscore(game_id)
fox_nba_odds(game_id); fox_nba_teams()
fox_nba_team_roster(team_id)
fox_nba_team_stats(team_id)
fox_nba_team_gamelog(team_id)
fox_nba_standings(team_id)
fox_nba_league_leaders()
```

### CROSSWALKS (KEYED ON ESPN_TEAM_ID)

```
nba_team_crosswalk(season = 2024)
# ESPN · NBA Stats · Fox ids, one row per team
nba_schedule_crosswalk(2024)
nba_player_crosswalk(2024)
load_nba_team_crosswalk(2024)  # release
load_nba_schedule_crosswalk()
load_nba_player_crosswalk()
```

### INJURIES, SALARIES, MOCK DRAFTS

```
rotowire_injuries(); bref_injuries()
spotrac_team_cap(season = 2025)   # cap sheets
hoopshype_salaries()              # salaries
nbadraft_mock_draft(year = 2026)  # mock draft
```

### OFFICIATING

```
# official.nba.com: L2M reports, ref crews
nba_l2m(game_id = "0042500405")
nba_l2m_games(season = 2025)   # end year
nba_referee_assignments("2025-06-05",
  league = "nba")          # or "gl", "wnba"
```

Typed empty schemas, so `bind_rows()` over many dates works.

### EUROLEAGUE · EUROLEAGUE_*

```
euroleague_games("E", "E2025")  # U = EuroCup
euroleague_game_pbp(game_code = 1,
                    season_code = "E2025")
```

15 wrappers: `competitions seasons rounds clubs people games game_stats game_report game_boxscore game_header game_points` (shots) `standings(kind =) player_stats team_stats(mode =)`.

### BASKETBALL-REFERENCE · BREF_*

```
bref_players_stats(2025, table = "per_game")
bref_teams_stats(2025, table = "per_game")
bref_player_game_log("jamesle01", 2025)
bref_team_roster(team = "LAL", season = 2025)
bref_standings(2025); bref_draft(2025)
bref_awards(2025); bref_player_bios("a")
```

`table` takes the page's table id (`per_game`, `totals`, `advanced`, ...). Polite scraping: cache results, do not loop aggressively.

### REALGM · REALGM_* (chromote + Chrome)

```
realgm_players(); realgm_teams()
realgm_coaches(); realgm_gms()
realgm_player_stats(season = 2025,
  stat_type = "Averages",
  season_type = "Regular_Season")
realgm_team_stats(); realgm_standings()
realgm_transactions(); realgm_salary_cap()
realgm_rookie_scale()
realgm_draft(2025); realgm_draft_prospects()
realgm_early_entry()
realgm_future_free_agents()
realgm_players_abroad()
realgm_individual_games()
realgm_individual_seasons()
```

RealGM sits behind a Cloudflare challenge; the headless-Chrome session is opened lazily on first call.

### PER-GAME HELPERS (DATA REPOS)

`helper_espn_nba_pbp` `helper_espn_nba_team_box` `helper_espn_nba_player_box` (and the `_mbb_` trio) wrap the raw ESPN summary into the loader schemas — what `hoopR-nba-data` / `hoopR-mbb-data` run nightly.

### INTERNATIONAL · 23 FAMILIES

```
fiba_eurobasket_schedule(season = 2025)
fiba_bcl_pbp(game_id, season = 2025)
acb_pbp(match_id = 105401)    # Liga ACB
kbl_schedule(from_date = "20251101",
             to_date = "20251102")
nbl_rosetta_matches(season = 2026)
genius_competitions(org = "ESAKE")
```

`fiba_` `genius_` `atrium_` `nbl_rosetta_` `scb_` (BBL) `feb_` `tbf_` `aba_` `acb_` `bleague_` `sportpress_` (BNXT) `cba_` `cebl_` `sportality_` (Basketligaen) `easl_` `ibba_` `kbl_` `kzs_` `lba_` `lkl_` `pleague_` `tpbl_` `vtb_`. 148 competition aliases: `fiba_{wc,oly,bcl,bcla,eurobasket,americup,asiacup,afrobasket,icup}_*`, `slb_` `bal_` `nznbl_` `esake_` `lnbm_` `lub_` (Genius), `lnb_` `lnbp_` `nbl1_` `lmb_` `lpb_` (Atrium), each with `_schedule` `_pbp` `_player_box` `_team_box` `_shots`. Names, args and columns match sportsdataverse-py; 1 s between requests to a host; proxy via `GENIUS_PROXY` / `CBA_PROXY`.

### ANALYTICS UTILITIES (VECTORISED)

```
nba_true_shooting_pct(pts, fga, fta)
nba_effective_fg_pct(fgm, fg3m, fga)
nba_ft_rate(fta, fga)
nba_possessions(fga, fta, oreb, tov)
nba_pace(poss, minutes)
nba_offensive_rating(pts, poss)
nba_net_rating(off_rating, def_rating)
nba_usage_rate(fga, fta, tov, minutes,
  team_fga, team_fta, team_tov, team_minutes)
nba_four_factors(fgm, fg3m, fga, fta, ftm,
  oreb, tov, opp_dreb)
nba_game_score(pts, fgm, fga, fta, ftm, oreb,
  dreb, stl, ast, blk, pf, tov)
box |> nba_add_advanced_metrics()  # adds all
nba_per_minutes(df, cols)
nba_per_possessions(df, cols)
```

Also `nba_defensive_rating` `nba_assist_pct` `nba_assist_to_turnover` `nba_turnover_pct` `nba_oreb_pct`. League-agnostic: works on ESPN or Stats-API box frames.

### POSSESSIONS, RAPM, SHOT ZONES

```
poss <- nba_possession_lineups("0022200001")
# off_player_1..5, def_player_1..5, points
rapm <- nba_rapm(poss)   # ridge o/d/total
nba_shot_zones("0022200001")   # FGA by zone
nba_tracking_aggregate(
  seasons = c("2022-23", "2023-24"),
  pt_measure_type = "Drives",
  per_mode = "Totals")
```

RAPM needs a full season of possessions (5-8k) to be meaningful; bind many `nba_possession_lineups()` frames first.

### RESULT-SET REDUCERS & TIDY WRAPPERS

```
sets <- nba_playercareerstats("2544")   # list
nba_bind_sets(sets, tag_column = "set_name")
nba_join_sets(sets, join_key = "PLAYER_ID")
nba_nest_sets(sets, keep_cols = "PLAYER_ID")
nba_tidy_boxscore(game_id = "0022200021")
nba_tidy_player_profile(player_id = "2544")
nba_tidy_franchise_history()
```

Media: `nba_player_headshot_url(player_id, size = "1040x760")` · `nba_team_logo_url(team_id, variant = "primary")`.

# NBA Stats API :: nba_* wrappers for stats.nba.com

Python siblings: nba_api · sportsdataverse.nba.nba_stats_* · WNBA: wehoop wnba_* · overview ?nba_stats

## CONVENTIONS

### CALLING PATTERN

```
library(hoopR)
sets <- nba_leaguedashplayerstats(
  season = year_to_season(2023), # "2023-24"
  season_type = "Regular Season",
  per_mode = "PerGame",
  league_id = "00")   # "20" = G-League
names(sets)             # result-set names
df <- sets$LeagueDashPlayerStats  # one tibble
nba_bind_sets(sets)     # or join_/nest_sets
```

Args mirror the endpoint's query string in snake_case; `season` defaults to the previous season. `?parameter_descriptions` (and the Parameter Descriptions article) lists all 300 with `api_name`, `valid_values`, `default`, `functions`. Game ids are 10-char strings (`"0022200021"`); player/team ids are character. Every wrapper returns a **named list** of result sets (V3 boxscores/pbp return nested-JSON-derived frames).

Headers + proxy: all calls go through `request_with_proxy()`; set `options(hoopR.proxy = ...)` or pass `proxy =` when stats.nba.com blocks your IP (cloud / datacenter ranges often hang). Go slow: `Sys.sleep()` between calls.

### PLAY-BY-PLAY & LIVE

| | |
|---|---|
| `nba_pbp(game_id, on_court = TRUE, version = "v3")` | V2/V3 pbp + on-court lineups |
| `nba_pbps(game_ids, nest_data = FALSE)` | multi-game pbp |
| `nba_playbyplayv3` `nba_data_pbp` | raw V3 · data.nba.com feed |
| `nba_live_pbp` `nba_live_boxscore` | CDN live feeds |
| `nba_winprobabilitypbp` `nba_gamerotation` | WP overlay · rotations |
| `nba_todays_scoreboard` `nba_scoreboard` `_scoreboardv2` `_scoreboardv3` | slates |
| `nba_schedule` `nba_scheduleleaguev2int` | season schedule |
| `nba_leaguegamefinder` `nba_leaguegamelog` | game finder / log |

### SHOT CHARTS · SYNERGY · TRACKING

`nba_shotchartdetail` `nba_shotchartleaguewide` `nba_shotchartlineupdetail` `nba_synergyplaytypes` `nba_leaguedashptstats` `nba_leaguedashptdefend` `nba_leaguedashptteamdefend` `nba_leaguedashplayerptshot` `nba_leaguedashteamptshot` `nba_leaguedashoppptshot` `nba_shot_zones` `nba_tracking_aggregate` (page 3). The three `*ptshot` wrappers filter on `shot_dist_range`.

### BOX SCORES

| V2 (result sets) | V3 (nested) |
|---|---|
| `nba_boxscoretraditionalv2` | `nba_boxscoretraditionalv3` |
| `nba_boxscoreadvancedv2` | `nba_boxscoreadvancedv3` |
| `nba_boxscorefourfactorsv2` | `nba_boxscorefourfactorsv3` |
| `nba_boxscoremiscv2` | `nba_boxscoremiscv3` |
| `nba_boxscorescoringv2` | `nba_boxscorescoringv3` |
| `nba_boxscoreusagev2` | `nba_boxscoreusagev3` |
| `nba_boxscoreplayertrackv2` | `nba_boxscoreplayertrackv3` |
| `nba_boxscorematchups` | `nba_boxscorematchupsv3` |
| `nba_boxscoresummaryv2` | `nba_boxscoresummaryv3` |
| `nba_boxscoredefensive` `_defensivev2` | `nba_boxscorehustlev2` |
| `nba_hustlestatsboxscore` | `nba_boxscoresimilarityscore` `nba_glalumboxscoresimilarityscore` |

### LEAGUE DASHBOARDS

| Players | Teams |
|---|---|
| `nba_leaguedashplayerstats` | `nba_leaguedashteamstats` |
| `nba_leaguedashplayerclutch` | `nba_leaguedashteamclutch` |
| `nba_leaguedashplayershotlocations` | `nba_leaguedashteamshotlocations` |
| `nba_leaguedashplayerbiostats` | `nba_leaguedashlineups` |
| `nba_leaguehustlestatsplayer` `_playerleaders` | `nba_leaguehustlestatsteam` `_teamleaders` |
| `nba_leagueplayerondetails` | `nba_leaguelineupviz` |
| `nba_leagueseasonmatchups` `nba_matchupsrollup` | `nba_leaguestandings` `_standingsv3` `nba_iststandings` |

### LEADERS & HOMEPAGE

`nba_leagueleaders` `nba_alltimeleadersgrids` `nba_homepageleaders` `nba_homepagev2` `nba_leaderstiles` `nba_assistleaders` `nba_assisttracker` `nba_dunkscoreleaders` `nba_gravityleaders` `nba_defensehub` `nba_playoffpicture` `nba_commonplayoffseries` `nba_fantasywidget` `nba_infographicfanduelplayer`.

### PLAYERS

```
nba_commonallplayers("2023-24")   # id lookup
nba_playerindex(); nba_player_dict()
nba_commonplayerinfo(player_id = "2544")
nba_playercareerstats(player_id = "2544",
                      per_mode = "Totals")
nba_playergamelog("2544", season = "2023-24")
nba_playerprofilev2(player_id = "2544")
```

**Dashboards ( nba_playerdashboardby* )**

`clutch` `gamesplits` `generalsplits` `lastngames` `opponent` `shootingsplits` `teamperformance` `yearoveryear`

**Tracking ( nba_playerdashpt* )**

`pass` `reb` `shotdefend` `shots`

**More**

`nba_playergamelogs` `nba_playergamestreakfinder` `nba_playernextngames` `nba_playervsplayer` `nba_playercompare` `nba_playerawards` `nba_playerestimatedmetrics` `nba_playerfantasyprofile` `nba_playerfantasyprofilebargraph` `nba_playercareerbycollege` `nba_playercareerbycollegerollup` `nba_playerheadshot` `nba_cumestatsplayer` `nba_cumestatsplayergames`

`nba_tidy_player_profile(player_id)` reduces the profile sets to one frame.

### TEAMS

```
nba_teams(); nba_team_dict()
nba_teaminfocommon(team_id = "1610612747")
nba_commonteamroster(team_id, "2023-24")
nba_teamgamelog(team_id, season = "2023-24")
nba_teamyearbyyearstats(team_id)
```

**Dashboards ( nba_teamdashboardby* )**

`clutch` `gamesplits` `generalsplits` `lastngames` `opponent` `shootingsplits` `teamperformance` `yearoveryear`

**Tracking / lineups**

`nba_teamdashptpass` `nba_teamdashptreb` `nba_teamdashptshots` `nba_teamdashlineups`

**More**

`nba_teamdetails` `nba_teamestimatedmetrics` `nba_teamgamelogs` `nba_teamgamestreakfinder` `nba_teamhistoricalleaders` `nba_teamplayerdashboard` `nba_teamplayeronoffdetails` `nba_teamplayeronoffsummary` `nba_teamvsplayer` `nba_teamandplayersvsplayers` `nba_commonteamyears` `nba_cumestatsteam` `nba_cumestatsteamgames`

### DRAFT & COMBINE

`nba_drafthistory` `nba_draftboard` `nba_draftcombinestats` `nba_draftcombineplayeranthro` `nba_draftcombinedrillresults` `nba_draftcombinespotshooting` `nba_draftcombinenonstationaryshooting`.

### FRANCHISES

`nba_franchisehistory` `nba_franchiseplayers` `nba_franchiseleaders` `nba_franchiseleaderswrank` · `nba_tidy_franchise_history()` = active + defunct rows in one frame.

### VIDEO

`nba_videodetails` `nba_videodetailsasset` `nba_videoevents` `nba_videoeventsasset` `nba_videostatus`.

### COMMON FILTERS

| Arg | Typical values |
|---|---|
| `season` | `"2023-24"` — use `year_to_season(2023)` |
| `season_type` | `"Regular Season"` `"Playoffs"` `"Pre Season"` `"PlayIn"` |
| `per_mode` | `"Totals"` `"PerGame"` `"Per100Possessions"` `"Per36"` |
| `measure_type` | `"Base"` `"Advanced"` `"Misc"` `"Four Factors"` `"Scoring"` `"Opponent"` `"Usage"` `"Defense"` |
| `league_id` | `"00"` NBA · `"10"` WNBA · `"20"` G-League |
| `player_or_team` | `"P"` / `"T"` (synergy, game finder) |
| `date_from` `date_to` | `"MM/DD/YYYY"` or `""` |
| `last_n_games` `month` `opponent_team_id` `period` | `0` = all (hustle: `last_n_games` deprecated, ignored) |
| `start_period` `end_period` | V3 boxscore / pbp slicing |

### PIPELINE SKETCH

```
library(dplyr); library(purrr)
ids <- nba_schedule("2023-24")$game_id
pbp <- nba_pbps(game_ids = ids[1:20])
poss <- ids[1:20] |>
  map(nba_possession_lineups) |> bind_rows()
nba_rapm(poss) |> arrange(desc(rapm))
```

Real RAPM needs a whole season — see page 3.

# hoopR :: CHEAT SHEET

Men's basketball data in R: NBA, NBA G-League, men's college basketball and 23 international league families plus EuroLeague / EuroCup. Tidy play-by-play, box scores, schedules and rosters from ESPN, a full `stats.nba.com` wrapper, KenPom / Torvik / CollegeBasketballData scrapers, Basketball-Reference, Fox and RealGM, plus season-level loaders from the SportsDataverse data releases. Women's sibling: **wehoop**; Python mirror: `sportsdataverse.nba` / `.mbb` in **sportsdataverse-py**. Page 1 of 4: install, surfaces, loaders, credentials, errors, helpers.

v3.2.0

R >= 4.1.0 · httr2 · data.table
hoopR.sportsdataverse.org

## GET STARTED

```
install.packages("hoopR")            # CRAN
pak::pak("sportsdataverse/hoopR")    # main
remotes::install_github(
  "sportsdataverse/hoopR")
library(hoopR)
```

1,316 exports (667 in the international families). Every function returns a tibble / data.frame (or a named list of tibbles for multi-table endpoints). Prefix tells you the source; page tells you where to look.

## SURFACES BY PREFIX

| Prefix | Source | Page |
|---|---|---|
| `load_nba_*` `load_mbb_*` `load_nba_stats_*` `load_ncaa_mbb_*` | season parquet/rds releases (zero scraping) | 2 · 3 |
| `espn_nba_*` `espn_mbb_*` | ESPN Site / Core APIs, live | 2 · 3 |
| `nba_*` | stats.nba.com Stats API (~160 endpoints) | 4 |
| `nbagl_*` | G-League (Stats API, league 20) | 3 |
| `nba_l2m*` `nba_referee_assignments` | official.nba.com L2M + referee crews | 3 |
| `euroleague_*` | EuroLeague / EuroCup APIs (keyless) | 3 |
| `fiba_*` `genius_*` `atrium_*` `acb_*` `kbl_*` ... | 23 international families + 148 competition aliases | 3 |
| `kp_*` | KenPom (subscription login) | 2 |
| `cbbd_*` | CollegeBasketballData API (key) | 2 |
| `torvik_*` | Bart Torvik T-Rank | 2 |
| `ncaa_mbb_*` | NCAA.com NET + teams | 2 |
| `fox_nba_*` `fox_mbb_*` | Fox Sports (Bifrost) | 2 · 3 |
| `bref_*` | Basketball-Reference | 3 |
| `realgm_*` | RealGM (needs chromote + Chrome) | 3 |
| `spotrac_` `hoopshype_` `nbadraft_` `rotowire_` | salaries · mock draft · injuries | 3 |
| `*_crosswalk` | ESPN ⇄ NBA Stats / KenPom / Torvik / Fox ids · MBB team names | 2 · 3 |

## DATA LOADERS (FAST PATH)

Full seasons from the `sportsdataverse-data` GitHub releases; each loader takes exactly the seasons its release publishes. Seasons are the **end year** (2024 = 2023-24). ESPN join ids load as integer.

```
pbp  <- load_nba_pbp(seasons = 2022:2024)
tb   <- load_nba_team_box(
          seasons = most_recent_nba_season())
pb   <- load_mbb_player_box(seasons = 2024)
sch  <- load_mbb_schedule(seasons = 2020:2024)
# stream straight into a DB table
load_nba_pbp(2024, dbConnection = con,
             tablename = "nba_pbp")
```

| Dataset | `load_nba_*` | `load_mbb_*` |
|---|---|---|
| play-by-play | `pbp` | `pbp` |
| box scores | `team_box` `player_box` | `team_box` `player_box` |
| schedules | `schedule` | `schedule` |
| standings | `standings` | `standings` |
| rosters | `rosters` `game_rosters` | `rosters` `game_rosters` |
| season stats | `player_stats` `team_stats` | `player_stats` `team_stats` |
| people | `player_core` `officials` `draft` | `player_core` `officials` |
| crosswalks | `team_` `schedule_` `player_crosswalk` | same three |
| shots / value | `shots` `player_impact` | `shots` `ratings` `player_value` |
| conferences | `groups` `group_seasons` `group_aliases` `team_group_seasons` | same four |

Also `load_nba_stats_*` — 18 stats.nba.com-derived release loaders (schedule pbp possessions lineups shots rosters boxscores game_logs season_stats leaguedash standings officials coaches draft) + `update_nba_stats_db()` — and `load_ncaa_mbb_*` — 13 stats.ncaa.org loaders incl. `rapm` / `rapm_within_team`. Group loaders keep each season's own conference alignment.

**Local play-by-play database**

```
update_nba_db(dbdir = ".",
              dbname = "hoopR_db",
              tblname = "hoopR_nba_pbp")
update_mbb_db()    # RSQLite / DBI (Suggests)
con <- DBI::dbConnect(RSQLite::SQLite(),
                      "hoopR_db")
```

## LIVE ESPN + STATS API

```
# ESPN: game_id from espn_*_scoreboard
#       or from load_*_schedule
espn_nba_scoreboard(20230423)   # YYYYMMDD
espn_nba_pbp(game_id = 401071880)
espn_mbb_game_all(401479672)    # pbp + boxes
# stats.nba.com: 10-char game ids, "2023-24"
nba_pbp("0022200021", on_court = TRUE)
nba_leaguedashplayerstats(
  season = year_to_season(2023))
```

ESPN wrappers take `game_id`, `team_id`, `athlete_id`, `season` / `year`. Stats API wrappers mirror the endpoint's query params (snake_case) and return a **named list** of result sets. Live `espn_*_pbp` plays carry the releases' clock, period and timeout columns.

## CREDENTIALS & OPTIONS

```
# KenPom (active subscription)  ?kp_user_pw
Sys.setenv(KP_USER = "you@example.com",
           KP_PW = "...")
has_kp_user_and_pw()   # TRUE before kp_*()
# CollegeBasketballData  ?register_cbbd
Sys.setenv(CBBD_API_KEY = "XXXX-YOUR-KEY")
has_cbbd_key(); cbbd_key()
# or persist both in ~/.Renviron
usethis::edit_r_environ()
# session-wide proxy (stats.nba.com / ESPN)
options(hoopR.proxy = "http://host:port")
options(hoopR.fox_data_key = "...")  # Fox
```

Stats API requests go through `request_with_proxy()`, which sets the required `x-nba-stats-*` / Referer headers; `hoopR.proxy`, `http_proxy` env or an explicit `proxy=` arg are honoured. `progressively()` + `progressr` give progress bars on multi-season loads.

## ERRORS & DEFAULTS

```
r <- tryCatch(acb_pbp(match_id = 105401),
  hoopR_no_data = function(e) NULL)
# 404, nothing there  -> hoopR_no_data
# request failed      -> hoopR_asset_fetch
# 400 / 422           -> hoopR_bad_request
```

International, EuroLeague and officiating wrappers raise classed conditions (all inherit `hoopR_error`; EuroLeague and officiating use `hoopR_fetch_error`, EuroLeague `hoopR_invalid_request`) instead of returning an empty frame. Stats API `season` now defaults to the previous season, `year_to_season(most_recent_nba_season() - 2)`: pass it for the current one.

## HELPERS

| | |
|---|---|
| `most_recent_nba_season()` | current NBA season end year |
| `most_recent_mbb_season()` | current college season end year |
| `year_to_season(2023)` | `"2023-24"` for Stats API |
| `csv_from_url(...)` | fast csv reader behind loaders |
| `progressively(f, p)` | wrap `f` with a progressor |
| `login()` | KenPom session (reads `KP_USER` / `KP_PW`) |
| `hoopR:::request_with_proxy()` | httr2 GET w/ NBA headers + proxy (internal) |
| `hoopR:::rds_from_url()` `:::check_status()` | internal readers / status guard |
| `teams_links` | bundled KenPom team + conference names by year (2002-27, 9,044 rows) |
| `mbb_team_names` | NCAA / 247 / Sports-Reference / Warren Nolan names, 368 MBB teams |
| `parameter_descriptions` | all 300 Stats API params, values + defaults |
| `nba_player_dict()` `nba_team_dict()` | Stats-API id dictionaries |
| `nba_player_headshot_url()` `nba_team_logo_url()` | CDN media URLs |

## OVERVIEW TOPICS (?NAME)

`?hoopR_loaders` · `?espn_basketball` · `?nba_stats` · `?kp` · `?register_cbbd` · `?espn_basketball_player_core`. pkgdown reference groups follow the same split.

## WIDER SPORTSDATAVERSE

| | |
|---|---|
| **wehoop** | WNBA + women's college, same API shape |
| **bigballR** | stats.ncaa.org MBB lineups / on-off |
| **cfbfastR · baseballr · fastRhockey** | sister sport packages |
| **sportsdataverse-py** | `sportsdataverse.nba` (`nba_stats_*`, `espn_nba_*`, `load_nba_*`) · `sportsdataverse.mbb` (`espn_mbb_*`, `ncaa_mbb_*`) |
| **nba_api** | Python stats.nba.com reference |
| data repos | `sportsdataverse-data` (loaders) · `hoopR-nba-data` · `hoopR-mbb-data` · `hoopR-nba-stats-data` |

# Men's college basketball :: loaders · ESPN · KenPom · CBD · Torvik

Women's mirror: **wehoop** (`espn_wbb_*`, `load_wbb_*`) · stats.ncaa.org lineups: **bigballR** · Python: `sportsdataverse.mbb`

## LOADERS + ESPN · ESPN_MBB_*

### SEASON LOADERS

```
library(hoopR)
pbp <- load_mbb_pbp(seasons = 2022:2024)
tb  <- load_mbb_team_box(seasons = 2024)
pb  <- load_mbb_player_box(seasons = 2024)
sch <- load_mbb_schedule(seasons = 2006:2024)
load_mbb_standings(); load_mbb_rosters()
load_mbb_player_stats(); load_mbb_team_stats()
load_mbb_ratings(); load_mbb_shots()
load_mbb_player_value()
load_ncaa_mbb_{pbp,possessions,lineups,shots,
  rapm,team_box,player_box,rosters,...}()
# ^ stats.ncaa.org releases
update_mbb_db(dbdir = ".",
              tblname = "hoopR_mbb_pbp")
```

Releases: `espn_mens_college_basketball_*` tags on `sportsdataverse-data`; pbp / shots 2003 and 2006+, rosters 2025+.

### GAMES (LIVE ESPN)

```
espn_mbb_scoreboard("20221117")   # day slate
espn_mbb_pbp(game_id = 401479672)
espn_mbb_team_box(401479672)
espn_mbb_player_box(401479672)
espn_mbb_game_all(401479672)   # pbp + boxes
espn_mbb_wp(401256760)   # win prob per play
espn_mbb_betting(401256760)
espn_mbb_game_rosters(401256760)
```

Core-API per-game extras (event id): `espn_mbb_game_odds` `_game_predictor` `_game_probabilities` `_game_propbets` `_game_officials` `_game_broadcasts` `_game_situation` `_game_play` `_game_play_personnel` `_game_team_statistics` `_game_team_leaders` `_game_team_linescores` `_game_team_score` `_game_team_roster` `_game_powerindex`.

### TEAMS, PLAYERS, LEAGUE

```
espn_mbb_teams(); espn_mbb_conferences()
espn_mbb_rankings(); espn_mbb_standings(2021)
espn_mbb_team_stats(150, year = 2020)
espn_mbb_player_stats(athlete_id = 4593919,
                      year = 2021)
espn_mbb_team_roster(150, season = 2024)
espn_mbb_team_schedule(150, season = 2024)
espn_mbb_player_gamelog(athlete_id, 2024)
```

Also: `espn_mbb_team_current_roster` `_team_record` `_team_injuries` `_team_leaders` `_team_news` `_team_season_statistics` `_player_overview` `_player_splits` `_player_career_stats` `_player_eventlog` `_player_info` `_athletes_index` `_leaders` `_season_leaders` `_injuries` `_news` `_futures` `_powerindex` (BPI) `_tournaments` `_week_rankings` `_season_groups` `_coaches` `_venues` `_calendar`.

## KENPOM · KP_* (subscription)

```
Sys.setenv(KP_USER = "you@x.com",
           KP_PW = "***")     # ?kp_user_pw
kp_pomeroy_ratings(2020, 2024)   # years
kp_efficiency(2020, 2024)
kp_fourfactors(2020, 2024)
kp_team_schedule("Florida St.", year = 2022)
kp_team_players(team = "Duke", year = 2024)
kp_playerstats(metric = "eFG", year = 2024)
kp_box(game_id, year)
kp_winprob(game_id, year)
```

| Family | Functions |
|---|---|
| Ratings | `kp_pomeroy_ratings` `kp_pomeroy_archive_ratings` `kp_program_ratings` `kp_trends` `kp_hca` `kp_kpoy` |
| Team stats | `kp_efficiency` `kp_fourfactors` `kp_teamstats` `kp_pointdist` `kp_height` `kp_foul_trouble` `kp_playerstats` |
| Team pages | `kp_team_schedule` `kp_team_players` `kp_team_player_stats` `kp_team_history` `kp_team_lineups` `kp_team_depth_chart` `kp_gameplan` `kp_opptracker` `kp_minutes_matrix` `kp_coach_history` |
| Games | `kp_box` `kp_winprob` `kp_fanmatch(date)` `kp_game_attrs(year, attr)` |
| Conf / misc | `kp_conf(year, conf)` `kp_confhistory` `kp_confstats` `kp_arenas` `kp_officials` `kp_referee` `kp_player_career` |
| Auth | `login()` `has_kp_user_and_pw()` `kp_user_email()` `kp_password()` |

Team names must match KenPom spelling: `teams_links` lists them, with conferences, by year (2002-27). Team pages build their link from the name.

## BART TORVIK · TORVIK_*

```
torvik_ratings(year = 2024)   # T-Rank
torvik_team_factors(2024)
torvik_player_stats(2024)
torvik_game_schedule(2024)
torvik_game_stats(2024)
torvik_ncaa_results(2015, 2024, type = "team")
```

## CONFERENCE HISTORY (GROUPS)

```
load_mbb_team_group_seasons(2002:2025)
load_mbb_groups(); load_mbb_group_seasons()
load_mbb_group_aliases()  # ESPN/NCAA/KP
```

Membership as it was each season, not today's alignment; one `group_id` survives renames.

## COLLEGEBASKETBALLDATA · CBBD_*

```
# key: ?register_cbbd
Sys.setenv(CBBD_API_KEY = "XXXX")
cbbd_games(season = 2024, team = "Duke")
cbbd_plays_game(game_id,
  shooting_plays_only = FALSE)
cbbd_lines(season = 2024, conference = "ACC")
cbbd_ratings_adjusted(season = 2024)
cbbd_stats_player_season(2024, team = "Duke")
```

| Family | Functions |
|---|---|
| Games | `cbbd_games` `cbbd_games_teams` `cbbd_games_players` `cbbd_games_media` `cbbd_scoreboard` |
| Plays | `cbbd_plays_game` `cbbd_plays_date` `cbbd_plays_team` `cbbd_plays_player` `cbbd_plays_tournament` `cbbd_play_types` |
| Lineups / subs | `cbbd_lineups_game` `cbbd_lineups_team` `cbbd_substitutions_game` `cbbd_substitutions_team` `cbbd_substitutions_player` |
| Stats | `cbbd_stats_team_season` `cbbd_stats_team_shooting_season` `cbbd_stats_team_leaderboard` `cbbd_stats_player_season` `cbbd_stats_player_shooting_season` |
| Ratings / polls | `cbbd_ratings_adjusted` `cbbd_ratings_srs` `cbbd_ratings_elo` `cbbd_rankings` |
| Lines | `cbbd_lines` `cbbd_lines_providers` |
| Teams / venues | `cbbd_teams` `cbbd_teams_roster` `cbbd_teams_directory` `cbbd_teams_season_overview` `cbbd_conferences` `cbbd_conferences_history` `cbbd_venues` |
| Recruiting / draft | `cbbd_recruiting_players` `cbbd_recruiting_teams` `cbbd_recruiting_portal` `cbbd_draft_picks` `cbbd_draft_teams` `cbbd_draft_positions` |
| Key | `cbbd_key()` `has_cbbd_key()` `check_cbbd_key()` |

Most take `season`, `team`, `conference`, `start_date_range` / `end_date_range`, `season_type` filters.

## NCAA & FOX

```
ncaa_mbb_NET_rankings()   # current NET
ncaa_mbb_teams(year = 2024, division = 1)
fox_mbb_pbp(game_id)
fox_mbb_boxscore(game_id)
fox_mbb_odds(game_id); fox_mbb_teams()
fox_mbb_teams_all(); fox_mbb_league_leaders()
fox_mbb_team_roster(team_id)
fox_mbb_team_stats(team_id)
fox_mbb_team_gamelog(team_id)
fox_mbb_standings(team_id)
```

Fox ids differ from ESPN ids — bridge with the crosswalks below.

## CROSSWALKS (KEYED ON ESPN_TEAM_ID)

```
mbb_team_crosswalk(season = 2024)
# ESPN · KenPom · Torvik · Fox ids per team
mbb_schedule_crosswalk(season = 2024)
mbb_player_crosswalk(season = 2024)
load_mbb_team_crosswalk(2024)   # release
load_mbb_schedule_crosswalk()
load_mbb_player_crosswalk()
```

The team crosswalk adds six name columns (`ncaa_team` `espn_pbp_name` `warren_nolan_team` `team_247` `sref_slug` `sref_team`) from the bundled `mbb_team_names` (368 teams).

## TYPICAL PIPELINE

```
library(dplyr)
pbp <- load_mbb_pbp(2024)
shots <- pbp |>
  filter(shooting_play,
         !is.na(coordinate_x)) |>
  count(team_id, score_value)
kp <- kp_pomeroy_ratings(2024, 2024)
xw <- load_mbb_team_crosswalk(2024)
tb <- load_mbb_team_box(2024) |>
  left_join(xw,
    by = c("team_id" = "espn_team_id"))
```

Column names above (`shooting_play`, `coordinate_x`, `score_value`) follow the ESPN pbp schema documented in `?load_mbb_pbp`.

## ELSEWHERE

Per-game ESPN helpers used by the data repo builders: `helper_espn_mbb_pbp` `_team_box` `_player_box`. Python parity: `sportsdataverse.mbb.espn_mbb_pbp`, `load_mbb_pbp`, `ncaa_mbb_*` (bigballR port).

# NBA, G-League & international :: ESPN · EuroLeague · FIBA

stats.nba.com `nba_*` wrappers: page 4 · WNBA mirror: **wehoop** · Python: `sportsdataverse.nba` / `nbagl`

## LOADERS + ESPN · ESPN_NBA_*

### SEASON LOADERS (1994+)

```
library(hoopR)
pbp <- load_nba_pbp(seasons = 2022:2024)
tb  <- load_nba_team_box(2024)
pb  <- load_nba_player_box(2024)
sch <- load_nba_schedule(2002:2024)
load_nba_standings(); load_nba_rosters()
load_nba_player_stats(); load_nba_team_stats()
load_nba_draft(); load_nba_player_core()
update_nba_db(dbdir = ".",
              tblname = "hoopR_nba_pbp")
```

Releases: `espn_nba_*` tags on `sportsdataverse-data`. Box scores, schedule, player stats / core 1994+; pbp, shots, standings 2002+; rosters 2025+.

### GAMES (LIVE ESPN)

```
espn_nba_scoreboard(20230423)   # YYYYMMDD
espn_nba_pbp(game_id = 401071880)
espn_nba_team_box(401071880)
espn_nba_player_box(401071880)
espn_nba_game_all(401283399)    # pbp + boxes
espn_nba_wp(401283399)
espn_nba_betting(401283399)
espn_nba_game_rosters(401283399)
```

Core-API per-event extras: `espn_nba_game_odds` `_game_predictor` `_game_probabilities` `_game_propbets` `_game_officials` `_game_broadcasts` `_game_situation` `_game_play` `_game_play_personnel` `_game_team_statistics` `_game_team_leaders` `_game_team_linescores` `_game_team_records` `_game_powerindex`. Live pbp carries the release's clock, period and timeout columns.

### TEAMS, PLAYERS, LEAGUE

```
espn_nba_teams(); espn_nba_standings(2021)
espn_nba_team_stats(team_id = 13, year = 2020)
espn_nba_player_stats(athlete_id = 1966,
                      year = 2022)
espn_nba_team_roster(13)
espn_nba_team_schedule(13, season = 2024)
espn_nba_team_depthchart(team_id = 13)
espn_nba_draft(season = 2024)
espn_nba_transactions(season = 2024)
espn_nba_freeagents(); espn_nba_injuries()
espn_nba_futures()
espn_nba_player_contracts(athlete_id = 1966)
espn_nba_powerindex(season = 2024)   # BPI
```

Also: `espn_nba_athletes_index` `_player_overview` `_player_gamelog` `_player_splits` `_player_career_stats` `_player_awards` `_player_contract` `_team_record` `_team_injuries` `_team_leaders` `_team_news` `_season_draft` `_draft_athletes` `_draft_rounds` `_coaches` `_franchises` `_venues` `_calendar` `_leaders` `_news`.

## G-LEAGUE · NBAGL_*

```
nbagl_schedule(season = year_to_season(2023))
nbagl_standings("2023-24")   # league 20
nbagl_players()
nbagl_pbp(game_id, on_court = TRUE)
nbagl_live_pbp(game_id)
nbagl_live_boxscore(game_id)
```

Same Stats-API backend as `nba_*` (page 4); most `nba_*` wrappers also accept `league_id = "20"`.

## FOX SPORTS · FOX_NBA_*

```
fox_nba_pbp(game_id)
fox_nba_boxscore(game_id)
fox_nba_odds(game_id); fox_nba_teams()
fox_nba_team_roster(team_id)
fox_nba_team_stats(team_id)
fox_nba_team_gamelog(team_id)
fox_nba_standings(team_id)
fox_nba_league_leaders()
```

## CROSSWALKS (KEYED ON ESPN_TEAM_ID)

```
nba_team_crosswalk(season = 2024)
# ESPN · NBA Stats · Fox ids, one row per team
nba_schedule_crosswalk(2024)
nba_player_crosswalk(2024)
load_nba_team_crosswalk(2024)  # release
load_nba_schedule_crosswalk()
load_nba_player_crosswalk()
```

## INJURIES, SALARIES, MOCK DRAFTS

```
rotowire_injuries(); bref_injuries()
spotrac_team_cap(season = 2025)   # cap sheets
hoopshype_salaries()              # salaries
nbadraft_mock_draft(year = 2026)  # mock draft
```

## OFFICIATING

```
# official.nba.com: L2M reports, ref crews
nba_l2m(game_id = "0042500405")
nba_l2m_games(season = 2025)   # end year
nba_referee_assignments("2025-06-05",
  league = "nba")          # or "gl", "wnba"
```

Typed empty schemas, so `bind_rows()` over many dates works.

## EUROLEAGUE · EUROLEAGUE_*

```
euroleague_games("E", "E2025")  # U = EuroCup
euroleague_game_pbp(game_code = 1,
                    season_code = "E2025")
```

15 wrappers: `competitions` `seasons` `rounds` `clubs` `people` `games` `game_stats` `game_report` `game_boxscore` `game_header` `game_points` (shots) `standings(kind =)` `player_stats` `team_stats(mode =)`.

## BASKETBALL-REFERENCE · BREF_*

```
bref_players_stats(2025, table = "per_game")
bref_teams_stats(2025, table = "per_game")
bref_player_game_log("jamesle01", 2025)
bref_team_roster(team = "LAL", season = 2025)
bref_standings(2025); bref_draft(2025)
bref_awards(2025); bref_player_bios("a")
```

`table` takes the page's table id (`per_game`, `totals`, `advanced`, ...). Polite scraping: cache results, do not loop aggressively.

## REALGM · REALGM_* (chromote + Chrome)

```
realgm_players(); realgm_teams()
realgm_coaches(); realgm_gms()
realgm_player_stats(season = 2025,
  stat_type = "Averages",
  season_type = "Regular_Season")
realgm_team_stats(); realgm_standings()
realgm_transactions(); realgm_salary_cap()
realgm_rookie_scale()
realgm_draft(2025); realgm_draft_prospects()
realgm_early_entry()
realgm_future_free_agents()
realgm_players_abroad()
realgm_individual_games()
realgm_individual_seasons()
```

RealGM sits behind a Cloudflare challenge; the headless-Chrome session is opened lazily on first call.

## PER-GAME HELPERS (DATA REPOS)

`helper_espn_nba_pbp` `helper_espn_nba_team_box` `helper_espn_nba_player_box` (and the `_mbb_` trio) wrap the raw ESPN summary into the loader schemas — what `hoopR-nba-data` / `hoopR-mbb-data` run nightly.

## INTERNATIONAL · 23 FAMILIES

```
fiba_eurobasket_schedule(season = 2025)
fiba_bcl_pbp(game_id, season = 2025)
acb_pbp(match_id = 105401)    # Liga ACB
kbl_schedule(from_date = "20251101",
             to_date = "20251102")
nbl_rosetta_matches(season = 2026)
genius_competitions(org = "ESAKE")
```

`fiba_` `genius_` `atrium_` `nbl_rosetta_` `scb_` (BBL) `feb_` `tbf_` `aba_` `acb_` `bleague_` `sportpress_` (BNXT) `cba_` `cebl_` `sportality_` (Basketligaen) `easl_` `ibba_` `kbl_` `kzs_` `lba_` `lkl_` `pleague_` `tpbl_` `vtb_`. 148 competition aliases: `fiba_{wc,oly,bcl,bcla,eurobasket,americup,asiacup,afrobasket,icup}_*`, `slb_` `bal_` `nznbl_` `esake_` `lnbm_` `lub_` (Genius), `lnb_` `lnbp_` `nbl1_` `lmb_` `lpb_` (Atrium), each with `_schedule` `_pbp` `_player_box` `_team_box` `_shots`. Names, args and columns match sportsdataverse-py; 1 s between requests to a host; proxy via `GENIUS_PROXY` / `CBA_PROXY`.

## ANALYTICS UTILITIES (VECTORISED)

```
nba_true_shooting_pct(pts, fga, fta)
nba_effective_fg_pct(fgm, fg3m, fga)
nba_ft_rate(fta, fga)
nba_possessions(fga, fta, oreb, tov)
nba_pace(poss, minutes)
nba_offensive_rating(pts, poss)
nba_net_rating(off_rating, def_rating)
nba_usage_rate(fga, fta, tov, minutes,
  team_fga, team_fta, team_tov, team_minutes)
nba_four_factors(fgm, fg3m, fga, fta, ftm,
  oreb, tov, opp_dreb)
nba_game_score(pts, fgm, fga, fta, ftm, oreb,
  dreb, stl, ast, blk, pf, tov)
box |> nba_add_advanced_metrics()  # adds all
nba_per_minutes(df, cols)
nba_per_possessions(df, cols)
```

Also `nba_defensive_rating` `nba_assist_pct` `nba_assist_to_turnover` `nba_turnover_pct` `nba_oreb_pct`. League-agnostic: works on ESPN or Stats-API box frames.

## POSSESSIONS, RAPM, SHOT ZONES

```
poss <- nba_possession_lineups("0022200001")
# off_player_1..5, def_player_1..5, points
rapm <- nba_rapm(poss)   # ridge o/d/total
nba_shot_zones("0022200001")   # FGA by zone
nba_tracking_aggregate(
  seasons = c("2022-23", "2023-24"),
  pt_measure_type = "Drives",
  per_mode = "Totals")
```

RAPM needs a full season of possessions (5-8k) to be meaningful; bind many `nba_possession_lineups()` frames first.

## RESULT-SET REDUCERS & TIDY WRAPPERS

```
sets <- nba_playercareerstats("2544")   # list
nba_bind_sets(sets, tag_column = "set_name")
nba_join_sets(sets, join_key = "PLAYER_ID")
nba_nest_sets(sets, keep_cols = "PLAYER_ID")
nba_tidy_boxscore(game_id = "0022200021")
nba_tidy_player_profile(player_id = "2544")
nba_tidy_franchise_history()
```

Media: `nba_player_headshot_url(player_id, size = "1040x760")` · `nba_team_logo_url(team_id, variant = "primary")`.

# NBA Stats API :: nba_* wrappers for stats.nba.com

Python siblings: **nba_api** · `sportsdataverse.nba.nba_stats_*` · WNBA: **wehoop** `wnba_*` · overview `?nba_stats`

## CONVENTIONS

### CALLING PATTERN

```
library(hoopR)
sets <- nba_leaguedashplayerstats(
  season = year_to_season(2023), # "2023-24"
  season_type = "Regular Season",
  per_mode = "PerGame",
  league_id = "00")   # "20" = G-League
names(sets)              # result-set names
df <- sets$LeagueDashPlayerStats  # one tibble
nba_bind_sets(sets)      # or join_/nest_sets
```

Args mirror the endpoint's query string in snake_case; `season` defaults to the previous season. `?parameter_descriptions` (and the Parameter Descriptions article) lists all 300 with `api_name`, `valid_values`, `default`, `functions`. Game ids are 10-char strings (`"0022200021"`); player/team ids are character. Every wrapper returns a **named list** of result sets (V3 boxscores/pbp return nested-JSON-derived frames).

Headers + proxy: all calls go through `request_with_proxy()`; set `options(hoopR.proxy = ...)` or pass `proxy =` when stats.nba.com blocks your IP (cloud / datacenter ranges often hang). Go slow: `Sys.sleep()` between calls.

### PLAY-BY-PLAY & LIVE

| | |
|---|---|
| `nba_pbp(game_id, on_court = TRUE, version = "v3")` | V2/V3 pbp + on-court lineups |
| `nba_pbps(game_ids, nest_data = FALSE)` | multi-game pbp |
| `nba_playbyplayv3` `nba_data_pbp` | raw V3 · data.nba.com feed |
| `nba_live_pbp` `nba_live_boxscore` | CDN live feeds |
| `nba_winprobabilitypbp` `nba_gamerotation` | WP overlay · rotations |
| `nba_todays_scoreboard` `nba_scoreboard` `_scoreboardv2` `_scoreboardv3` | slates |
| `nba_schedule` `nba_scheduleleaguev2int` | season schedule |
| `nba_leaguegamefinder` `nba_leaguegamelog` | game finder / log |

### SHOT CHARTS · SYNERGY · TRACKING

`nba_shotchartdetail` `nba_shotchartleaguewide` `nba_shotchartlineupdetail` `nba_synergyplaytypes` `nba_leaguedashptstats` `nba_leaguedashptdefend` `nba_leaguedashptteamdefend` `nba_leaguedashplayerptshot` `nba_leaguedashteamptshot` `nba_leaguedashoppptshot` `nba_shot_zones` `nba_tracking_aggregate` (page 3). The three `*ptshot` wrappers filter on `shot_dist_range`.

## BOX SCORES

| V2 (result sets) | V3 (nested) |
|---|---|
| `nba_boxscore traditionalv2` | `nba_boxscoretraditionalv3` |
| `nba_boxscore advancedv2` | `nba_boxscoreadvancedv3` |
| `nba_boxscore fourfactorsv2` | `nba_boxscorefourfactorsv3` |
| `nba_boxscoremiscv2` | `nba_boxscoremiscv3` |
| `nba_boxscore scoringv2` | `nba_boxscorescoringv3` |
| `nba_boxscoreusagev2` | `nba_boxscoreusagev3` |
| `nba_boxscore playertrackv2` | `nba_boxscoreplayertrackv3` |
| `nba_boxscore matchups` | `nba_boxscorematchupsv3` |
| `nba_boxscore summaryv2` | `nba_boxscoresummaryv3` |
| `nba_boxscore defensive _defensivev2` | `nba_boxscorehustlev2` |
| `nba_hustlestats boxscore` | `nba_boxscore similarityscore` `nba_glalum boxscoresimilarityscore` |

## LEAGUE DASHBOARDS

| Players | Teams |
|---|---|
| `nba_leaguedash playerstats` | `nba_leaguedash teamstats` |
| `nba_leaguedash playerclutch` | `nba_leaguedash teamclutch` |
| `nba_leaguedash playershotlocations` | `nba_leaguedash teamshotlocations` |
| `nba_leaguedash playerbiostats` | `nba_leaguedashlineups` |
| `nba_leaguehustlestats player` `_playerleaders` | `nba_leaguehustlestats team` `_teamleaders` |
| `nba_league playerondetails` | `nba_leaguelineupviz` |
| `nba_league seasonmatchups` `nba_matchupsrollup` | `nba_leaguestandings` `_standingsv3` `nba_iststandings` |

## LEADERS & HOMEPAGE

`nba_leagueleaders` `nba_alltimeleadersgrids` `nba_homepageleaders` `nba_homepagev2` `nba_leaderstiles` `nba_assistleaders` `nba_assisttracker` `nba_dunkscoreleaders` `nba_gravityleaders` `nba_defensehub` `nba_playoffpicture` `nba_commonplayoffseries` `nba_fantasywidget` `nba_infographicfanduelplayer`.

## PLAYERS

```
nba_commonallplayers("2023-24")   # id lookup
nba_playerindex(); nba_player_dict()
nba_commonplayerinfo(player_id = "2544")
nba_playercareerstats(player_id = "2544",
                      per_mode = "Totals")
nba_playergamelog("2544", season = "2023-24")
nba_playerprofilev2(player_id = "2544")
```

**Dashboards (`nba_playerdashboardby*`)**

`clutch` `gamesplits` `generalsplits` `lastngames` `opponent` `shootingsplits` `teamperformance` `yearoveryear`

**Tracking (`nba_playerdashpt*`)**

`pass` `reb` `shotdefend` `shots`

**More**

`nba_playergamelogs` `nba_playergamestreakfinder` `nba_playernextngames` `nba_playervsplayer` `nba_playercompare` `nba_playerawards` `nba_playerestimatedmetrics` `nba_playerfantasyprofile` `nba_playerfantasyprofilebargraph` `nba_playercareerbycollege` `nba_playercareerbycollegerollup` `nba_playerheadshot` `nba_cumestatsplayer` `nba_cumestatsplayergames`

`nba_tidy_player_profile(player_id)` reduces the profile sets to one frame.

## TEAMS

```
nba_teams(); nba_team_dict()
nba_teaminfocommon(team_id = "1610612747")
nba_commonteamroster(team_id, "2023-24")
nba_teamgamelog(team_id, season = "2023-24")
nba_teamyearbyyearstats(team_id)
```

**Dashboards (`nba_teamdashboardby*`)**

`clutch` `gamesplits` `generalsplits` `lastngames` `opponent` `shootingsplits` `teamperformance` `yearoveryear`

**Tracking / lineups**

`nba_teamdashptpass` `nba_teamdashptreb` `nba_teamdashptshots` `nba_teamdashlineups`

**More**

`nba_teamdetails` `nba_teamestimatedmetrics` `nba_teamgamelogs` `nba_teamgamestreakfinder` `nba_teamhistoricalleaders` `nba_teamplayerdashboard` `nba_teamplayeronoffdetails` `nba_teamplayeronoffsummary` `nba_teamvsplayer` `nba_teamandplayersvsplayers` `nba_commonteamyears` `nba_cumestatsteam` `nba_cumestatsteamgames`

## DRAFT & COMBINE

`nba_drafthistory` `nba_draftboard` `nba_draftcombinestats` `nba_draftcombineplayeranthro` `nba_draftcombinedrillresults` `nba_draftcombinespotshooting` `nba_draftcombinenonstationaryshooting`.

## FRANCHISES

`nba_franchisehistory` `nba_franchiseplayers` `nba_franchiseleaders` `nba_franchiseleaderswrank` · `nba_tidy_franchise_history()` = active + defunct rows in one frame.

## VIDEO

`nba_videodetails` `nba_videodetailsasset` `nba_videoevents` `nba_videoeventsasset` `nba_videostatus`.

## COMMON FILTERS

| Arg | Typical values |
|---|---|
| `season` | `"2023-24"` — use `year_to_season(2023)` |
| `season_type` | `"Regular Season"` `"Playoffs"` `"Pre Season"` `"PlayIn"` |
| `per_mode` | `"Totals"` `"PerGame"` `"Per100Possessions"` `"Per36"` |
| `measure_type` | `"Base"` `"Advanced"` `"Misc"` `"Four Factors"` `"Scoring"` `"Opponent"` `"Usage"` `"Defense"` |
| `league_id` | `"00"` NBA · `"10"` WNBA · `"20"` G-League |
| `player_or_team` | `"P"` / `"T"` (synergy, game finder) |
| `date_from` `date_to` | `"MM/DD/YYYY"` or `""` |
| `last_n_games` `month` `opponent_team_id` `period` | `0` = all (hustle: `last_n_games` deprecated, ignored) |
| `start_period` `end_period` | V3 boxscore / pbp slicing |

## PIPELINE SKETCH

```
library(dplyr); library(purrr)
ids <- nba_schedule("2023-24")$game_id
pbp <- nba_pbps(game_ids = ids[1:20])
poss <- ids[1:20] |>
  map(nba_possession_lineups) |> bind_rows()
nba_rapm(poss) |> arrange(desc(rapm))
```

Real RAPM needs a whole season — see page 3.